

Most People Are Energy Blind when it comes to AI Predictions

Posted by u/WorldyBridges33 • • 0 points • 178 comments

Speculations on AI continuing to develop infinitely are interesting, but they are energy blind. There is a 0.99 R squared correlation between GDP and energy consumption. As energy consumption increases, the economy grows, and even with the advent of renewables, total energy consumption has increased every year (with the exception of global economic recessions in 2008 and 2020). Technology without energy is just a sculpture.

AI requires massive amounts of energy (mostly generated through finite fuels like natural gas and uranium) to work. It also requires massive amounts of mostly finite materials, like copper, lithium, oil-derived plastics, and other rare earth metals for the datacenters. Once we reach the peak production of these finite energy/material supplies, the economy (as measured by GDP) will start to degrow.

Remember, the only reason that 90% of the population no longer works in farming is because of heavy diesel-powered farming equipment. Global oil production likely peaked in Q4 of 2018 (see here: <https://www.resilience.org/stories/2023-01-18/arthur-berman-peak-oil-the-hedonic-adjustment/>). As less and less oil is available each year, more people will slowly have to return to farming. As less and less natural gas is available each year, a smaller amount will be left over for electrical generation (AI) because a fixed amount of natural gas must be set aside for fertilizer production (Haber-Bosch process).

Also, further complicating this is the fact that the bulk of AI is developed in the USA right now, but a huge portion of remaining fossil fuels/rare earth metals are in countries that are antagonistic to the USA (like Russia and Iran).

You could say that renewables would solve this, but wind and solar rely heavily on fossil fuels for their creation. Wind turbines require 80 gallons of oil a year for lubrication, and the wind blades themselves are made from oil derivatives. Solar panels require coal to smelt

quarts for their production. They also rely on concrete which is derived from fossil fuels. Furthermore, they are intermittent, or they can only produce energy when the sun is shining/wind is strong.

Even if batteries become more efficient, our energy consumption will only increase due to Jevons Paradox (see here: [https://en.wikipedia.org/wiki/Jevons_paradox](https://en.wikipedia.org/wiki/Jevons%20paradox)), hastening the depletion of finite energy stores.

No one is thinking about the physical side of the AI story because they've grown accustomed to thinking about software as a non-physical entity. I think many people are in for a rude awakening when the realities of biophysical limits clash with the myth of infinite economic growth.

Comments

u/[deleted] • 1 points • 2024-10-11 07:00 UTC

[removed]

u/Accomplished-Big3997 • 1 points • 2023-12-02 14:57 UTC

Interesting, the question is can we get AI to help us with energy efficiency through the discovery of new materials and data evaluation and implementation before the we run out of recourses. I would think there is going to be some kind of understanding here soon that AI growth with be because of growing with us. Helping us achieve the technology we need to solve the energy problem for itself as it evolves. Devs will have to work very hard helping models understand how human needs intertwine with artificial intelligence means in a balanced harmonious way.

u/Artificial_Lives • 1 points • 2023-11-29 16:57 UTC

I think op has a point if we are talking 100+ or more years and assuming processing doesn't get more efficient and assuming we don't make new energy sources lol.

Small modular reactors are going to be huge a d they're coming back to places like the USA as well

u/Talosian_cagecleaner • 1 points • 2023-11-27 23:35 UTC

What a strange economy. You don't account for usage outside the case when it comes to computing. We will need those materials whether or not pursue AI. If there are efficiency gains, that even further incentivizes it.

Bang for buck becomes an easy answer. That's the point AI is waiting for. It's not there yet except as a proof of concept.

u/JoeStrout • 1 points • 2023-11-27 19:12 UTC

Those materials you mention are abundant in the solar system. If we need them that badly, we'll go get them.

u/sinfulmonk2 • 1 points • 2023-11-27 17:54 UTC

Did we forget quantum computing? Adding to some of the comments such energy production scaling up and other alternatives emerging, there is also the fact that quantum computers are gonna get better and are gonna give unimaginable computational advantage over current computers that AI models currently run on. The energy efficiency and computing speed of quantum computers is mind bending.